

DRIFA-Net Project — Current Implementation Status

Detailed handoff for local NVIDIA T1000 implementation with Claude Code

Purpose. This document records the current project state, dataset substitutions, notebook changes, architecture status, training configuration, the Colab crash issue, and the recommended local implementation path.

1. Project Objective

The project is an implementation/adaptation of the DRIFA-Net research-paper GitHub repository. The immediate goal is to get the main pipeline running with **HAM10000 + Brain Tumor MRI**, rather than reproducing every experiment in the paper. Dropout/uncertainty experiments are intentionally postponed.

Required pipeline: dataset loading → preprocessing/augmentation → dual CNN branches → MFA → RGSA → repeated MIFA fusion → feature concatenation → global pooling → two classification heads → training → evaluation.

2. Original Repository Assumptions

The main notebook being adapted is **Code.ipynb**. Its original dataset-loading code expected preprocessed NumPy files:

```
X_train_HAM10000_ISIC_2018.npy
y_train_HAM10000_ISIC_2018.npy
X_test_HAM10000_ISIC_2018.npy
y_test_HAM10000_ISIC_2018.npy
```

Those files were not available, so raw Kaggle datasets are now loaded directly and converted into arrays expected by the downstream network. The original second dataset was SIPaKMeD; it has been replaced with Brain Tumor MRI.

3. Current Datasets

Branch	Dataset	Classes	Verified split
s / input 1	Brain Tumor MRI	4	5,600 train / 1,600 test
h / input 2	HAM10000	7	8,012 train / 2,003 test

HAM10000 path

```
/content/drive/MyDrive/Jio Institute/Term 3/Projects/Computer Vision/DRIFA/Datasets/HAM10000
```

Contains HAM10000_metadata.csv, HAM10000_images_part_1, HAM10000_images_part_2, plus hmnist CSV files. The implementation uses metadata + JPG images.

Brain Tumor path

```
/content/drive/MyDrive/Jio Institute/Term 3/Projects/Computer Vision/DRIFA/Datasets/brain tumor
```

Contains Training/{glioma, meningioma, notumor, pituitary} and Testing/{glioma, meningioma, notumor, pituitary}.

4. HAM10000 Loading Changes

The missing .npy loading section was replaced with a raw-image loader. It reads metadata, finds each JPG in either image-part directory, resizes to 128×128 RGB, normalizes by 255, performs an 80/20 stratified split, and one-hot encodes labels.

```
akiec=0, bcc=1, bkl=2, df=3, mel=4, nv=5, vasc=6
```

Verified: X_train_h=(8012,128,128,3), y_train_h=(8012,7), X_test_h=(2003,128,128,3), y_test_h=(2003,7).

5. Brain Tumor Loading Changes

The SIPaKMeD loader was replaced by a folder-based Brain MRI loader. Existing Training/Testing folders are used directly; no extra train_test_split is required.

```
glioma=0, meningioma=1, notumor=2, pituitary=3
```

Verified before augmentation: X_train_s=(5600,128,128,3), y_train_s=(5600,4), X_test_s=(1600,128,128,3), y_test_s=(1600,4).

6. Brain MRI Augmentation

The repository's rotation/translation augmentation was retained. Each of the 5,600 training images receives a +20° rotation and a (5,5) translation, creating 11,200 augmented samples. To stay close to the original notebook, 4,773 augmented samples were selected and concatenated with the 5,600 originals, producing 10,373 Brain MRI training samples.

A Brain MRI test copy was also tripled and reduced to 2,003 samples so it can be paired with the 2,003-image HAM10000 test subset. The original 1,600-image X_test_s remains intact.

7. Architecture Status

Module	Status	Dataset change
MFA / DeeperGlobalLocalAttentionLayer1	Kept unchanged	None
MIFA / DeeperGlobalLocalAttentionLayer	Kept unchanged	None
RGSA	Kept unchanged	None
residual_GLC_branch1	Kept unchanged	None
Final classifier	Modified	Dense(5) → Dense(4); Dense(7) retained

MFA

MFA was left unchanged. It contains hierarchical/global attention and local channel information attention. Internal units such as 64 are feature dimensions, not class counts.

MIFA

MIFA was left unchanged. It performs multimodal global/local information fusion using min/average/max statistics from both branches.

RGSA

RGSA is a residual convolutional attention block using 3×3 convolutions, DeeperAttentionLayer1, BatchNormalization, ReLU and shortcut projection.

residual_GLC_branch1

The two branches progress through 64 → 128 → 256 → 512 feature channels, repeatedly applying RGSA/MFA and exchanging information through MIFA. These are feature widths, not class counts.

The network contains Dropout(0.25) with training=True, labelled MCD in the original code. These dropout layers were deliberately retained because they are part of the repository architecture. The separate dropout/uncertainty tests are postponed.

8. Final Model Adaptation

The original classifier was two heads: Dense(5) and Dense(7). It was changed to Dense(4) and Dense(7).

```
Input 1: Brain MRI → output 1 → Dense(4, softmax)
Input 2: HAM10000 → output 2 → Dense(7, softmax)
```

The model successfully built after importing GlobalAveragePooling2D. The shared vector after GlobalAveragePooling2D was reported as (None, 1024).

9. Training Configuration

Optimizer: Adam, learning rate 0.001. Both outputs use categorical_crossentropy. initial_gamma=0.5, giving loss_weights=[0.5,0.5].

Callbacks: ModelCheckpoint to best_DRIFA_brain_HAM.keras; EarlyStopping patience=100; ReduceLROnPlateau factor=0.2, patience=5, minimum learning rate 1e-5.

```
history = model.fit(
[X_train_s_aligned, X_train_h_aligned],
[y_train_s_aligned, y_train_h_aligned],
epochs=200,
validation_split=0.2,
shuffle=True,
```

```
callbacks=callbacks
)
```

10. Critical Sample-Alignment Issue

After augmentation, Brain MRI had 10,373 training samples while HAM10000 had 8,012. A multi-input Keras model requires the paired inputs and targets to have the same sample count. An alignment step therefore selected 8,012 Brain MRI samples to match HAM10000.

The previous alignment implementation created additional large NumPy copies. This contributed to severe memory pressure in Colab. For the local T1000 implementation, use memory-efficient indexing or preferably a `tf.data/generator` pipeline rather than retaining original, augmented and aligned copies simultaneously.

11. Why Colab Crashed

The session repeatedly crashed when training began. The likely issue was memory pressure: $128 \times 128 \times 3$ float32 arrays were duplicated for original Brain MRI, HAM10000, augmented Brain MRI, aligned copies and test copies, in addition to TensorFlow/model memory. A full process crash/restart can occur instead of a clean Python out-of-memory exception.

The T4 was correctly selected for GPU acceleration, but changing GPU type does not solve CPU/RAM duplication. The project is therefore being moved to a local NVIDIA T1000 with a cleaner memory strategy.

12. Current Local T1000 Strategy

- Use the same 128×128 RGB preprocessing and same model architecture.
- Do not retain unnecessary duplicate arrays; prefer `tf.data.Dataset`, generators, or memory-efficient indexing.
- Verify TensorFlow detects the NVIDIA T1000 before training.
- Start with batch size 4 or 8 and a 1–5 epoch smoke test.
- Only after the smoke test succeeds, run the intended longer training.
- Keep Brain MRI as input/output branch 1 and HAM10000 as input/output branch 2.
- Keep Dense(4) for Brain MRI and Dense(7) for HAM10000.
- Keep the repository Dropout(..., training=True) layers for now.
- Save checkpoints to a persistent local path.

13. Current End-to-End Pipeline

```
HAM10000 raw images + metadata
→ resize/normalize → stratified split → 7-class labels
Brain MRI Training/Testing folders
→ resize/normalize → 4-class labels
→ rotation + translation augmentation
Two CNN branches → MFA → RGSA → repeated MIFA → deeper RGSA/MFA
→ concatenate branch features → Dropout → GlobalAveragePooling2D
→ shared 1024-dimensional representation
■■■ Dense(4) → Brain MRI
■■■ Dense(7) → HAM10000
Total loss = 0.5 × Brain MRI CE + 0.5 × HAM10000 CE
```

14. What Has Been Changed vs Original

- Replaced missing HAM10000 .npz loads with direct raw-image loading.
- Replaced SIPaKMeD with Brain Tumor MRI.
- Skipped the original SIPaKMeD `train_test_split` because Brain MRI already provides Training/Testing folders.
- Kept HAM10000 at 7 classes.
- Changed the original 5-class output head to a 4-class Brain MRI head.
- Kept MFA, MIFA, RGSA and `residual_GLC_branch1` unchanged.
- Kept the original augmentation idea and selected 4,773 augmented Brain MRI samples.

- Added sample-count alignment because the two multimodal inputs must have equal training length.
- Identified memory duplication as the main reason Colab training crashed.
- Evaluation and prediction cells still need adaptation.

15. Pending Work

- Implement memory-efficient local data loading/alignment.
- Run a 1–5 epoch local T1000 smoke test.
- Verify both output losses and accuracies.
- Run full training after the smoke test.
- Adapt evaluation so both inputs have matching test sample counts.
- Replace binary thresholding such as `y_pred >= 0.5` with `np.argmax` for multiclass softmax outputs.
- Generate separate confusion matrices/classification reports for Brain MRI and HAM10000.
- Only after baseline training works, revisit dropout/uncertainty experiments.

16. Claude Code Handoff Prompt

Continue the DRIFA-Net GitHub implementation. We are adapting Code.ipynb from its original HAM10000 + SIPaKMeD setup to HAM10000 + Brain Tumor MRI.

Datasets:

- HAM10000: 10,015 images; 8,012 train / 2,003 test; 7 classes: akiec, bcc, bkl, df, mel, nv, vasc.
- Brain Tumor MRI: 7,200 images; 5,600 train / 1,600 test; 4 classes: glioma, meningioma, notumor, pituitary.

Preprocessing:

- Both resized to 128x128 RGB and normalized /255.
- HAM10000 uses metadata.csv and stratified 80/20 split.
- Brain MRI uses existing Training/Testing folders.
- Brain MRI augmentation: +20 degree rotation and translation (5,5).
- 4,773 augmented Brain MRI samples were selected and concatenated with 5,600 originals, producing 10,373 training samples.

Architecture:

- MFA unchanged.
- MIFA unchanged.
- RGSA unchanged.
- residual_GLC_branch1 unchanged.
- Final classifier changed from Dense(5)+Dense(7) to Dense(4)+Dense(7).
- Input/output order: Brain MRI first, HAM10000 second.
- GlobalAveragePooling2D shared vector was (None,1024).
- Dropout(0.25, training=True) remains because it is part of the original MCD architecture.

Training:

- Adam lr=0.001.
- categorical_crossentropy for both outputs.
- loss_weights=[0.5,0.5].
- ModelCheckpoint, EarlyStopping(patience=100), ReduceLROnPlateau(factor=.2, patience=5, min_lr=1e-5).
- Intended 200 epochs, but first run should be 1–5 epochs with batch_size 4 or 8.

Critical issue:

- Brain MRI training became 10,373 samples while HAM10000 has 8,012.
- Paired multi-input training requires equal sample counts.
- Previous alignment created duplicate NumPy arrays and caused Colab memory crashes.
- Implement alignment/data loading memory-efficiently using `tf.data/generators` or careful indexing.
- Local GPU is NVIDIA T1000.

Pending:

1. Verify GPU.
2. Implement memory-efficient training pipeline.
3. Smoke-test 1–5 epochs.
4. Full training.
5. Adapt evaluation to matching test arrays.
6. Use `argmax` for multiclass predictions, not binary thresholding.
7. Evaluate Brain MRI and HAM10000 separately.

Do not claim this is an exact reproduction of the paper: the second dataset has been changed from SIPaKMeD to

17. Important Scientific/Presentation Note

This is a repository adaptation, not yet a validated exact reproduction of the paper's experimental protocol. Replacing SIPaKMeD with Brain Tumor MRI changes the modality pair, class count, dataset statistics and potentially the interpretation of multimodal fusion. In a faculty presentation, describe it as an adaptation unless the original protocol is separately reproduced.

18. Immediate Next Step

On the local T1000: verify TensorFlow sees the GPU, implement the memory-efficient data pipeline, run a 1–5 epoch smoke test, verify both outputs train without a crash, then proceed to longer training and evaluation.